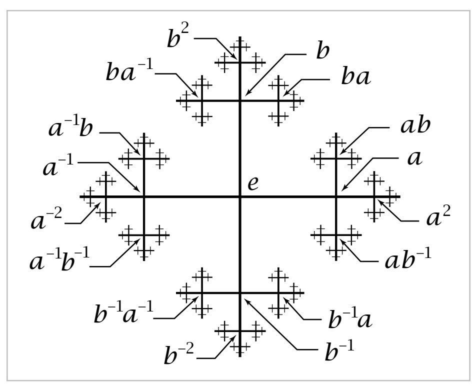
Cayley Complexes

Covering spaces can be used to describe a very classical method for viewing groups geometrically as graphs. Recall from Corollary 1.28 how we associated to each group presentation $G = \langle g_\alpha \mid r_\beta \rangle$ a 2-dimensional cell complex X_G with $\pi_1(X_G) \approx G$ by taking a wedge-sum of circles, one for each generator g_α, and then attaching a 2-cell for each relator r_β. We can construct a cell complex $\widetilde{X}_G$ with a covering space action of G such that $\widetilde{X}_G/G = X_G$ in the following way. Let the vertices of $\widetilde{X}_G$ be the elements of G themselves. Then, at each vertex $g \in G$, insert an edge joining g to gg_α for each of the chosen generators g_α. The resulting graph is known as the **Cayley graph** of G with respect to the generators g_α. This graph is connected since every element of G is a product of g_α's, so there is a path in the graph joining each vertex to the identity vertex e. Each relation r_β determines a loop in the graph, starting at any vertex g, and we attach a 2-cell for each such loop. The resulting cell complex $\widetilde{X}_G$ is the **Cayley complex** of G. The group G acts on $\widetilde{X}_G$ by multiplication on the left. Thus, an element $g \in G$ sends a vertex $g' \in G$ to the vertex gg', and the edge from g' to $g'g_\alpha$ is sent to the edge from gg' to $gg'g_\alpha$. The action extends to 2-cells in the obvious way. This is clearly a covering space action, and the orbit space is just X_G.

In fact $\widetilde{X}_G$ is the universal cover of X_G since it is simply-connected. This can be seen by considering the homomorphism $\varphi : \pi_1(X_G) \to G$ defined in the proof of Proposition 1.39. For an edge e_α in X_G corresponding to a generator g_α of G, it is clear from the definition of φ that $\varphi([e_\alpha]) = g_\alpha$, so φ is an isomorphism. In particular the kernel of φ, $p_*(\pi_1(\widetilde{X}_G))$, is zero, hence also $\pi_1(\widetilde{X}_G)$ since p_* is injective.

Let us look at some examples of Cayley complexes.

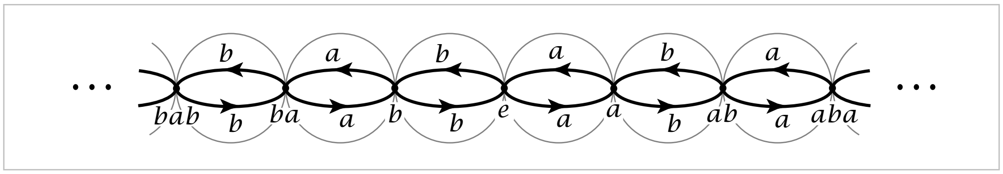
Example 1.45. When G is the free group on two generators a and b, X_G is $S^1 \vee S^1$ and $\widetilde{X}_G$ is the Cayley graph of $\mathbb{Z} * \mathbb{Z}$ pictured at the right. The action of a on this graph is a rightward shift along the central horizontal axis, while b acts by an upward shift along the central vertical axis. The composition ab of these two shifts then takes the vertex e to the vertex ab. Similarly, the action of any $w \in \mathbb{Z} * \mathbb{Z}$ takes e to the vertex w.



Example 1.46. The group $G = \mathbb{Z}\times\mathbb{Z}$ with presentation $\langle x, y \mid xyx^{-1}y^{-1} \rangle$ has X_G the torus $S^1 \times S^1$, and $\widetilde{X}_G$ is $\mathbb{R}^2$ with vertices the integer lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$ and edges the horizontal and vertical segments between these lattice points. The action of G is by translations $(x, y) \mapsto (x + m, y + n)$.

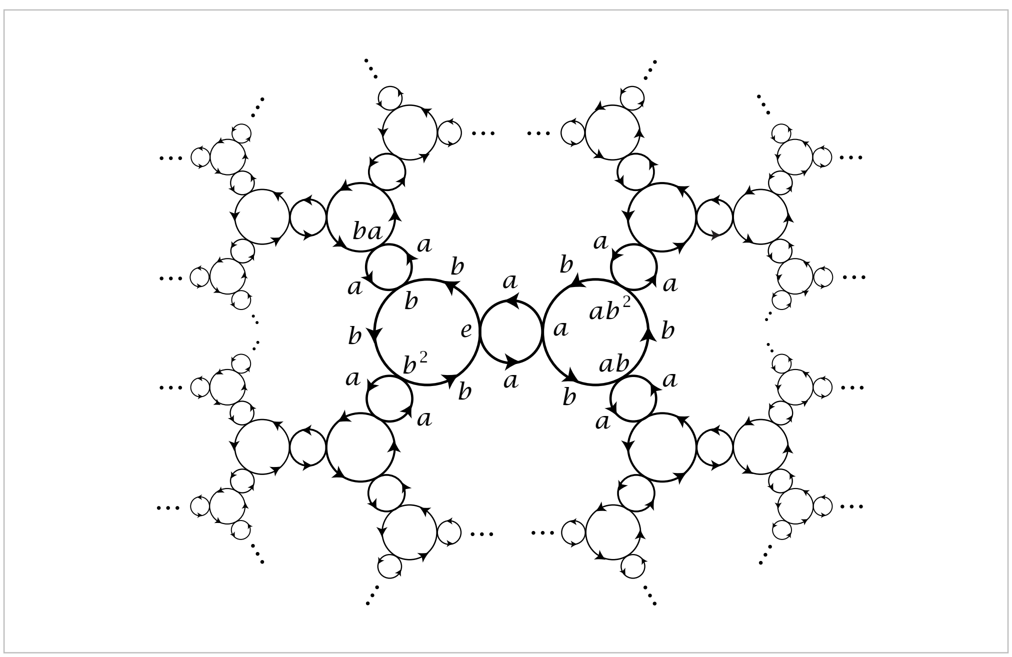
Example 1.47. For $G = \mathbb{Z}_2 = \langle\, x \mid x^2 \,\rangle$, X_G is $\mathbb{R}\mathrm{P}^2$ and $\widetilde{X}_G = S^2$. More generally, for $\mathbb{Z}_n = \langle\, x \mid x^n \,\rangle$, X_G is S^1 with a disk attached by the map $z \mapsto z^n$ and $\widetilde{X}_G$ consists of n disks $D_1, \cdots, D_n$ with their boundary circles identified. A generator of $\mathbb{Z}_n$ acts on this union of disks by sending D_i to D_{i+1} via a $2\pi/n$ rotation, the subscript i being taken mod n. The common boundary circle of the disks is rotated by $2\pi/n$.

Example 1.48. If $G = \mathbb{Z}_2 * \mathbb{Z}_2 = \langle\, a, b \mid a^2, b^2 \,\rangle$ then the Cayley graph is a union of an infinite sequence of circles each tangent to its two neighbors.



We obtain $\widetilde{X}_G$ from this graph by making each circle the equator of a 2-sphere, yielding an infinite sequence of tangent 2-spheres. Elements of the index-two normal subgroup $\mathbb{Z} \subset \mathbb{Z}_2 * \mathbb{Z}_2$ generated by ab act on $\widetilde{X}_G$ as translations by an even number of units, while each of the remaining elements of $\mathbb{Z}_2 * \mathbb{Z}_2$ acts as the antipodal map on one of the spheres and flips the whole chain of spheres end-for-end about this sphere. The orbit space X_G is $\mathbb{R}\mathrm{P}^2 \vee \mathbb{R}\mathrm{P}^2$.

It is not hard to see the generalization of this example to $\mathbb{Z}_m * \mathbb{Z}_n$ with the presentation $\langle\, a, b \mid a^m, b^n \,\rangle$, so that $\widetilde{X}_G$ consists of an infinite union of copies of the Cayley complexes for $\mathbb{Z}_m$ and $\mathbb{Z}_n$ constructed in Example 1.47, arranged in a tree-like pattern. The case of $\mathbb{Z}_2 * \mathbb{Z}_3$ is pictured below.



Exercises

1. For a covering space $p:\widetilde{X}\to X$ and a subspace $A\subset X$, let $\widetilde{A}=p^{-1}(A)$. Show that the restriction $p:\widetilde{A}\to A$ is a covering space.

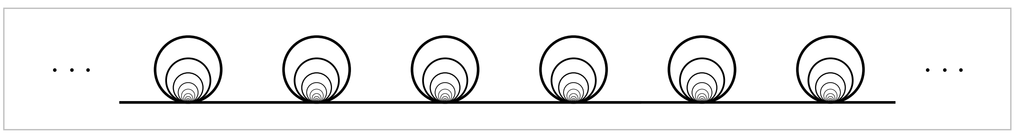
2. Show that if $p_1:\widetilde{X}_1\to X_1$ and $p_2:\widetilde{X}_2\to X_2$ are covering spaces, so is their product $p_1\times p_2:\widetilde{X}_1\times\widetilde{X}_2\to X_1\times X_2$.

3. Let $p:\widetilde{X}\to X$ be a covering space with $p^{-1}(x)$ finite and nonempty for all $x\in X$. Show that $\widetilde{X}$ is compact Hausdorff iff X is compact Hausdorff.

4. Construct a simply-connected covering space of the space $X\subset\mathbb{R}^3$ that is the union of a sphere and a diameter. Do the same when X is the union of a sphere and a circle intersecting it in two points.

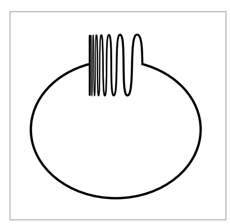
5. Let X be the subspace of $\mathbb{R}^2$ consisting of the four sides of the square $[0,1]\times[0,1]$ together with the segments of the vertical lines $x={}^1\!/_2,{}^1\!/_3,{}^1\!/_4,\cdots$ inside the square. Show that for every covering space $\widetilde{X}\to X$ there is some neighborhood of the left edge of X that lifts homeomorphically to $\widetilde{X}$. Deduce that X has no simply-connected covering space.

6. Let X be the shrinking wedge of circles in Example 1.25, and let $\widetilde{X}$ be its covering space shown in the figure below.



Construct a two-sheeted covering space $Y\to\widetilde{X}$ such that the composition $Y\to\widetilde{X}\to X$ of the two covering spaces is not a covering space. Note that a composition of two covering spaces does have the unique path lifting property, however.

7. Let Y be the *quasi-circle* shown in the figure, a closed subspace of $\mathbb{R}^2$ consisting of a portion of the graph of $y=\sin(1/x)$, the segment $[-1,1]$ in the y-axis, and an arc connecting these two pieces. Collapsing the segment of Y in the y-axis to a point gives a quotient map $f:Y\to S^1$. Show that f does not lift to the covering space $\mathbb{R}\to S^1$, even though $\pi_1(Y)=0$. Thus local path-connectedness of Y is a necessary hypothesis in the lifting criterion.



8. Let $\widetilde{X}$ and $\widetilde{Y}$ be simply-connected covering spaces of the path-connected, locally path-connected spaces X and Y. Show that if $X\simeq Y$ then $\widetilde{X}\simeq\widetilde{Y}$. [Exercise 10 in Chapter 0 may be helpful.]

9. Show that if a path-connected, locally path-connected space X has $\pi_1(X)$ finite, then every map $X\to S^1$ is nullhomotopic. [Use the covering space $\mathbb{R}\to S^1$.]

10. Find all the connected 2-sheeted and 3-sheeted covering spaces of $S^1\vee S^1$, up to isomorphism of covering spaces without basepoints.